

Efficient and compact TE–TM polarization converter built on silicon-on-insulator platform with a simple fabrication process

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An efficient TE–TM polarization converter built on a silicon-on-insulator nanophotonic platform is demonstrated. The strong cross-polarization coupling effect in air-cladded photonic-wire waveguides is employed to realize the conversion. A peak TE–TM coupling efficiency of 87% (−0.6 dB insertion loss) is measured experimentally. A polarization conversion efficiency of >92% with an overall insertion loss of < −1.6 dB is obtained in a wavelength range of 40 nm. The proposed device is compact, with a total length of 44 μm and can be fabricated with one lithography and etching step. © 2011 Optical Society of America

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Polarization control is one of the important issues in photonic integrated circuits [1]. Because of the geometry and the fabrication inaccuracy, the propagation constants, as well as some other characteristics, of the guided TE and TM modes are usually different. This kind of birefringence results in polarization-dependent dispersions or polarization-dependent losses for the integrated devices. This effect becomes more severe in high-index-contrast structures, e.g. photonic-wire waveguides on silicon-on-insulator (SOI) [2,3]. It is very difficult to build polarization-independent devices on such a platform [3–6]. In order to solve this problem, a polarization diversity scheme can be implemented [7–9]. In this case, the orthogonal polarization components of the input light are first split into two different waveguides. A polarization converter is then employed in one of the waveguides to rotate the polarization 90° so that only one polarization has to be processed by the rest of the chip. The same structure can be implemented at the output in order to combine the two polarizations. A polarization splitter can be easily implemented by using, e.g., a directional coupler [10]. However, a polarization converter is more difficult. Although different approaches have been introduced to realize such a device, they all need complex, or even nonstandard, fabrication steps, such as additional Si layer deposition, multistep lithography, etching with high alignment accuracy, nonvertical etching, etc. [11–19].

In this Letter, for the first time to our knowledge, we demonstrate a TE–TM polarization converter structure based on the cross-polarization coupling effect between SOI photonic-wire waveguides. It consists of only two parallel fully etched straight waveguides as shown in Fig. 1(a) and can be realized through a single lithography and etching step as needed for making the rest of the photonic circuits. This simple structure greatly facilitates the fabrication of such a device. We theoretically study and design the polarization converter. The optimized structure is also fabricated and characterized experimentally. A TE-to-TM polarization conversion efficiency

(PCE) of >92% with an overall insertion loss of < −1.6 dB (the lowest insertion loss reaches −0.6 dB) in a wavelength range of 40 nm is measured.

The working principle of the proposed polarization converter is based on the cross-polarization coupling in high-index-contrast SOI waveguides. It is known that the fundamental TE (E_x dominant) and TM (E_y dominant) polarized modes are orthogonal, and therefore no cross coupling between them is present, if the waveguide profile is symmetrical either horizontally or vertically. In the present design, the vertical symmetry is broken by using a different top-cladding material (air) from that of the buffer layer (silica), and the horizontal symmetry breaking is achieved by placing two parallel waveguides with different widths next to each other (cf. Fig. 1). Furthermore, if appropriate widths are chosen, phase matching conditions (i.e., the effective mode indices of

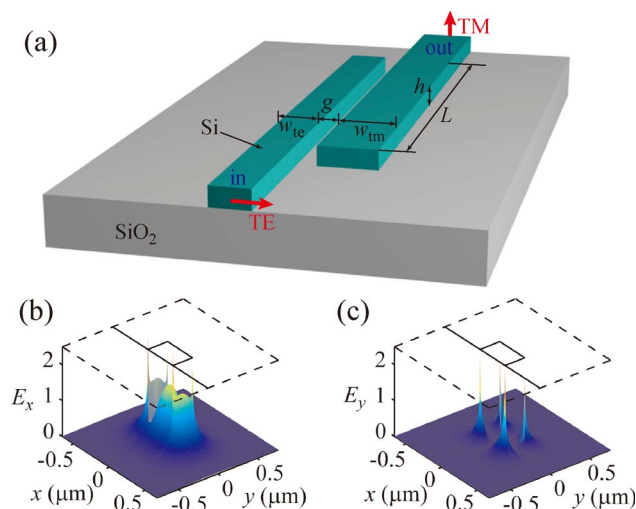


Fig. 1. (Color online) (a) Schematic structure of the proposed polarization converter. (b) E_x and (c) E_y field profiles (absolute value) of the fundamental TE mode in a single SOI wire waveguide of 500 nm width and 250 nm height with air cladding.

TE and TM modes are equal) can be fulfilled. Thus, efficient coupling ($\sim 100\%$) from the TE mode in one waveguide to the TM mode in the other waveguide can be achieved. This type of cross-polarization coupling is normally weak in a low-index-contrast waveguide or a structure without a significant symmetry breaking [14]. The resulting coupling length (defined as the length of the coupling section in order to achieve the maximal coupling efficiency) is very long, and therefore the whole structure becomes unrealistic to be implemented onchip. In the proposed device, this problem can be solved. Figures 1(b) and 1(c) show the mode fields of the fundamental TE mode in a single SOI wire waveguide with air cladding. One finds that there is still a significant nondominant polarization component (E_y). A similar phenomenon is found for the TM mode. This unique effect in the high-index-contrast structure greatly enhances the cross-polarization coupling, which makes the proposed device compact (i.e., the coupling length is kept within a few tens of micrometers) and practical to fabricate with the available tools.

The phase matching condition can be easily found by examining the mode properties of a single SOI wire with air cladding. Here, the thickness, h , of the top silicon layer of the SOI wafer is 250 nm, and we adopt $w_{\text{TE}} = 329$ nm, $w_{\text{TM}} = 550$ nm. The gap, g , is set to be 100 nm. We then use a commercial software (FIMMPROP, Photon Design, UK) employing an eigenmode expansion and matching method [20] to simulate the transmission property, T_{i-j} , of the present structure in three dimensions. T_{i-j} denotes the transmission coefficient from mode i at the in port [cf. Fig. 1(a)] to mode j at the out port, where i and j stand for TE or TM, and only the fundamental modes are considered here. Obviously, the TE-TM coupling efficiency, $T_{\text{TE-TM}}$, which is a key characteristic of the proposed converter, has to be maximized in the design. The PCE, defined as $C_{\text{TE-TM}} = T_{\text{TE-TM}} / (T_{\text{TE-TE}} + T_{\text{TE-TM}})$, which is a measure of the extinction ratio at the out port, is also an important figure [11,14,17]. Normally, a PCE better than 92% is required for some applications [11,14]. Because of reciprocity, this device can also be used in reverse (i.e., interchanging the in port and out port) for converting the TM mode to the TE mode. In this case, the coupling efficiency is expected to be the same, i.e., $T_{\text{TE-TM}}$, and the PCE can be defined accordingly as $C_{\text{TM-TE}} = T_{\text{TE-TM}} / (T_{\text{TM-TM}} + T_{\text{TE-TM}})$. The coupling efficiency $T_{\text{TE-TM}}$ is illustrated in Fig. 2(a) as a function of the coupling length, L , and the wavelength λ . Two contour lines of $C_{\text{TE-TM}}$ and $C_{\text{TM-TE}}$ at 92% are also plotted. The enclosed regions of both lines therefore denote the optimal parameter space concerning PCE. Figure 2(b) shows the wavelength response of $T_{\text{TE-TE}}$, $T_{\text{TM-TM}}$, and $T_{\text{TE-TM}}$ in decibel scale when $L = 44 \mu\text{m}$. The TE-TM coupling efficiency, $T_{\text{TE-TM}}$ reaches its maximum of 90.5% at $\lambda = 1530$ nm, which corresponds to an insertion loss of -0.44 dB. In a wavelength range of 71 nm (marked with gray), the PCEs and the insertion loss maintain better than 92% and -1.44 dB (1 dB band), respectively. One can also see in Fig. 2(a) that the peak coupling wavelength can be tuned by varying the coupling length, L , without compromising the performance significantly.

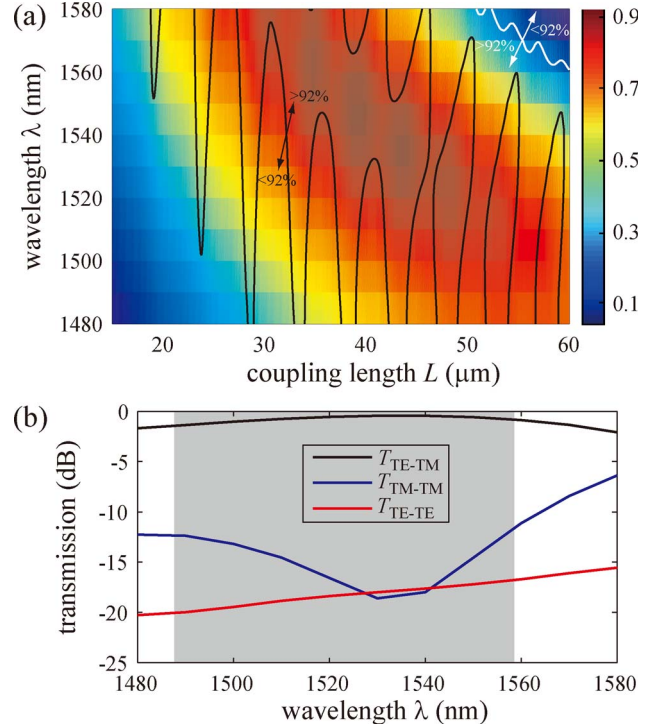


Fig. 2. (Color online) (a) Simulated transmission properties as a function of the coupling length, L , and the working wavelength, λ . The color map background shows $T_{\text{TE-TM}}$. The black and white contour curves show the PCEs at a level of 92% for $C_{\text{TE-TM}}$ and $C_{\text{TM-TE}}$, respectively. (b) Wavelength responses of $T_{\text{TE-TM}}$, $T_{\text{TM-TM}}$, and $T_{\text{TE-TE}}$ when $L = 44 \mu\text{m}$. Gray area indicates the wavelength range within which the PCEs are better than 92% and $T_{\text{TE-TM}}$ maintains better than -1 dB from its peak value. The rest of the structural parameters are: $w_{\text{TE}} = 329$ nm, $w_{\text{TM}} = 550$ nm, $g = 100$ nm, $h = 250$ nm.

The designed polarization converter was fabricated with standard SOI processing technology, including 100 keV electron-beam lithography (JEOL JBX-9300FS, JEOL, Japan) and fluorine-based inductively coupled-plasma reactive-ion-etching (STS Advanced Silicon Etcher, Surface Technology Systems, UK). Some pictures of the fabricated samples are shown in Fig. 3. Bended waveguide sections are attached to the idle ports of each waveguide to avoid any abrupt change of the index profile. The input and output waveguides are routed to the two cleaved edges of the sample, where light can be coupled in and out for measurement purposes. The measurement setup is sketched in Fig. 4(a). A tunable laser with a wavelength range of 1480 nm–1580 nm was used to characterize the fabricated devices. All the fibers used in the measurement were ordinary single-mode fibers. The polarization controllers (PC1 and PC2) were first

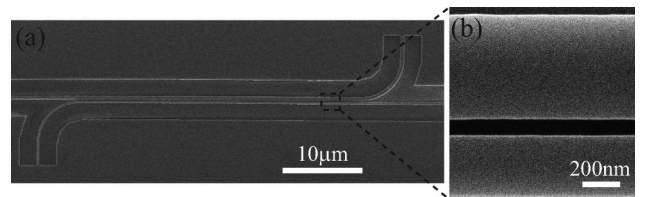


Fig. 3. Scanning electron microscope pictures of a fabricated device. (a) Full structure, (b) zoom-in view of the coupling section.

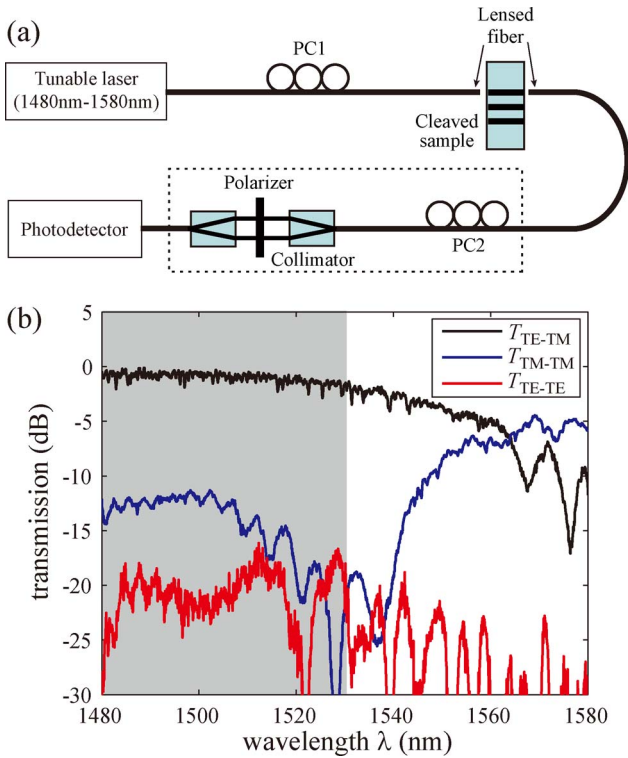


Fig. 4. (Color online) (a) Measurement setup for fabricated devices. (b) Normalized wavelength responses of the transmission T_{TE-TM} , T_{TM-TM} , and T_{TE-TE} . Gray area indicates the wavelength range within which the PCEs are better than 92% and T_{TE-TM} maintains better than -1 dB from its peak value.

calibrated using a reference sample with onchip polarizers and straight waveguides. The whole output path [within the dotted frame in Fig. 4(a)] were first bypassed by directly connecting the lensed fiber to the photodetector, and one onchip polarizer on the reference sample was aligned. The input polarization state was set by adjusting PC1 and monitoring the output power. Then, we moved to a straight waveguide, and the output path was inserted. PC2 was adjusted in order to align the output polarization plane to the polarizer. For the real samples, the polarization components of the output light can be selected by rotating the polarizer correspondingly.

Figure 4(b) shows the measured transmitted powers (normalized to those of a straight waveguide) between different polarizations. The overall loss of a reference straight waveguide is about -18 dB, including the coupling losses from the two cleaved facets, which can be improved by incorporating a spot-size converter [9]. They exhibit a good agreement with the simulation results shown in Fig. 2(b), except that the T_{TE-TM} curve is blue-shifted to a peak wavelength at 1500 nm. This probably resulted from a slight fabrication deviation from the design. The large oscillation in, e.g., the T_{TM-TM} curve, is likely caused by a slight error while setting the angle of the polarizer [10]. The peak TE-TM coupling efficiency reaches 87% (i.e., -0.6 dB insertion loss). The 1 dB coupling bandwidth is 40 nm (marked with gray), and

in this band the PCEs are also better than 92%. Theoretically, we found that the device performances are vulnerable to the variation of w_{TE} . The peak TE-TM coupling wavelength, as shown in Fig. 2(a), shifts with a rate of 25 nm/nm with respect to Δw_{TE} . This tight tolerance is expected for a directional-coupler-based device. More detailed analyses on the fabrication tolerance itself and methods to compensate the fabrication errors are undertaken.

In conclusion, we have demonstrated an efficient and compact polarization converter on SOI. The conversion is based on the cross-polarization coupling between two fully-etched photonic-wire waveguides. Unlike other similar devices, the fabrication of the proposed polarization converters does not need extra processing steps. We have experimentally obtained a PCE of >92% with an overall insertion loss of < -1.6 dB in a wavelength range of 40 nm in a 44- μ m-long device.

References

1. R. Alferness, IEEE J. Quantum Electron. **17**, 946 (1981).
2. D. Dai, Y. Shi, and S. He, Appl. Opt. **45**, 4941 (2006).
3. T. Y. L. Ang, S. T. Lim, S. Y. Lee, C. E. Png, and M. K. Chin, Appl. Opt. **48**, 2821 (2009).
4. D. Dai and S. He, Opt. Lett. **31**, 1988 (2006).
5. M.-M. Geng, L.-X. Jia, L. Zhang, Y.-L. Liu, L. Yang, and F. Li, Chin. Phys. Lett. **25**, 1333 (2008).
6. N. Zhu, J. Song, L. Wosinski, and S. He, IEEE Photonics Technol. Lett. **20**, 860 (2008).
7. T. Barwicz, M. R. Watts, M. A. Popović, P. T. Rakich, L. Socci, F. X. Kärtner, E. P. Ippen, and H. I. Smith, Nat. Photon. **1**, 57 (2007).
8. W. Bogaerts, D. Taillaert, P. Dumon, D. Van Thourhout, R. Baets, and E. Pluk, Opt. Express **15**, 1567 (2007).
9. H. Fukuda, K. Yamada, T. Tsuchizawa, T. Watanabe, H. Shinjima, and S. Itabashi, Opt. Express **16**, 4872 (2008).
10. H. Fukuda, K. Yamada, T. Tsuchizawa, T. Watanabe, S. Shinjima, and S. Itabashi, Opt. Express **14**, 12401 (2006).
11. H. Deng, D. O. Yevick, C. Brooks, and P. E. Jessop, J. Light-wave Technol. **23**, 432 (2005).
12. M. R. Watts and H. A. Haus, Opt. Lett. **30**, 138 (2005).
13. H. Fukuda, K. Yamada, T. Tsuchizawa, T. Watanabe, H. Shinjima, and S. Itabashi, Opt. Express **16**, 2628 (2008).
14. Z. Wang and D. Dai, J. Opt. Soc. Am. B **25**, 747 (2008).
15. Y. Yue, L. Zhang, M. Song, R. G. Beausoleil, and A. E. Willner, Opt. Express **17**, 20694 (2009).
16. K. Bayat, S. K. Chaudhuri, and S. Safavi-Naeini, Opt. Express **17**, 7145 (2009).
17. J. Pello, J. van der Tol, G. Roelkens, H. Ambrosius, and M. Smit, "Design of a new ultra-small polarization converter in InGaAsP/InP membrane," presented at the 15th European Conference on Integrated Optics, Cambridge, United Kingdom, April 2010, paper WeB2.
18. D. Vermeulen, S. Selvaraja, W. Bogaerts, and G. Roelkens, "High-efficiency broadband CMOS-compatible polarization rotator on SOI," presented at the Seventh International Conference on Group IV Photonics, Beijing, China, September 2010, paper WC6.
19. J. Zhang, M. Yu, G. Lo, and D. L. Kwong, IEEE J. Sel. Top. Quantum Electron. **16**, 53 (2010).
20. FIMMWAVE/FIMMPROP, Photon Design Ltd. <http://www.photond.com>.